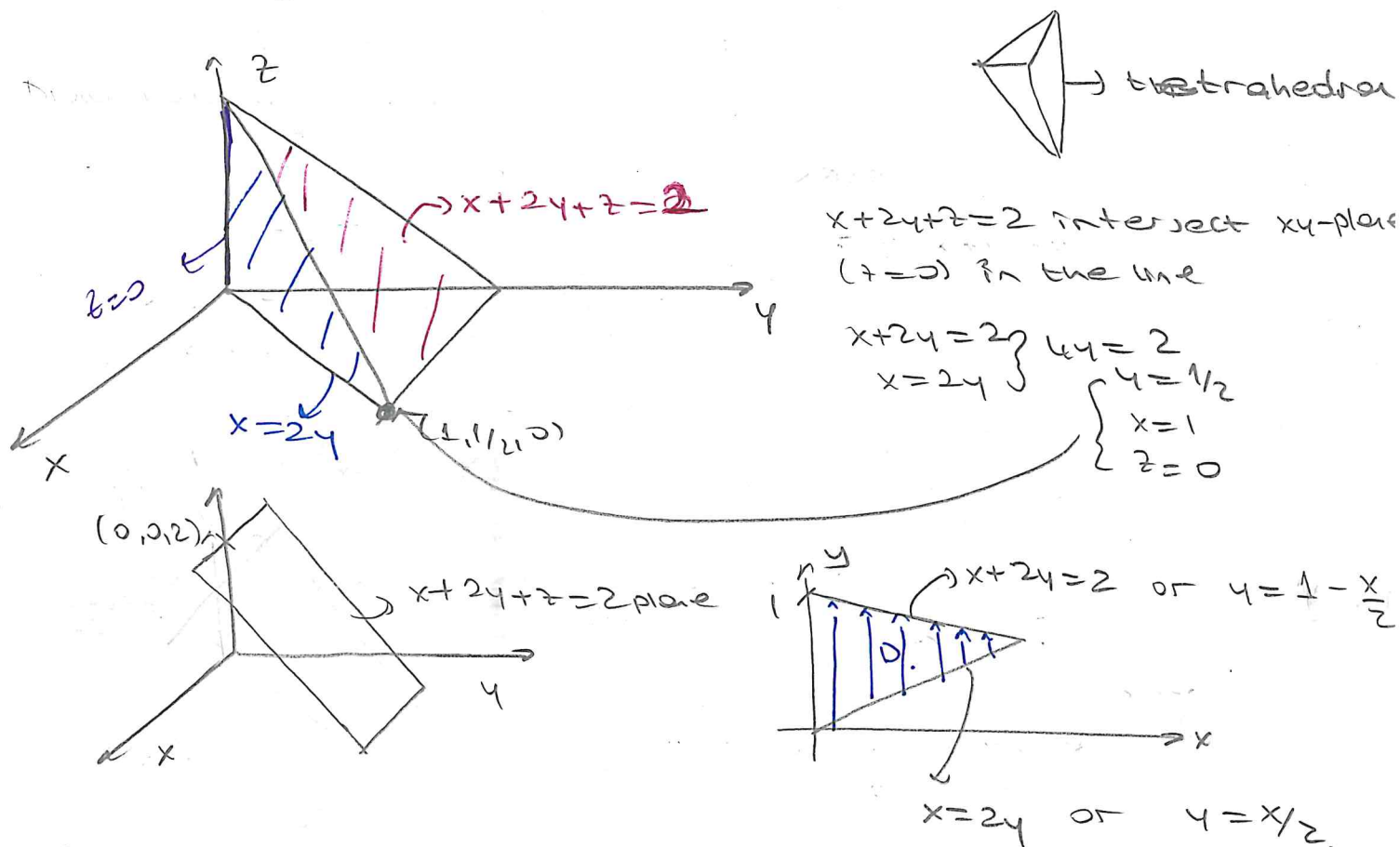


CALCULUS (II)

WEEK XIII

COMPUTING VOLUMES USING DOUBLE INTEGRALS

Ex 1 Find volume of the tetrahedron bounded by planes $x+2y+z=2$, $x=2y$, $x=0$ and $z=0$



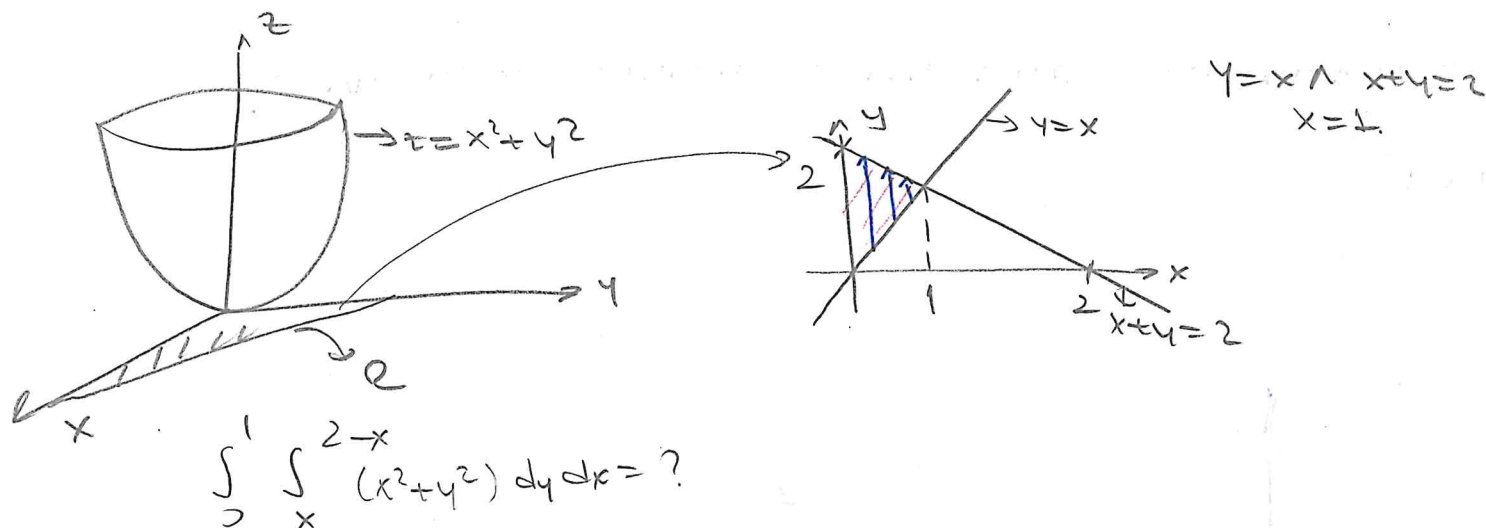
To sum up, the tetrahedron lies above the triangular region D in the xy plane bounded by the lines $x=2y$, $x+2y=2$ and $x=0$

$x+2y+z=2 \Rightarrow z=2-x-2y$. So, the required volume lies under the graph $z=2-x-2y$ and above

$$D = \{(x,y) \mid 0 \leq x \leq 1, \frac{x}{2} \leq y \leq 1-\frac{x}{2}\}$$

$$V = \iint_D (2-x-2y) dA = \int_0^1 \int_{x/2}^{1-x/2} (2-x-2y) dy dx = 1/3$$

Ex 2 Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the triangle enclosed by the lines $y = x, x = 2, x + y = 2$



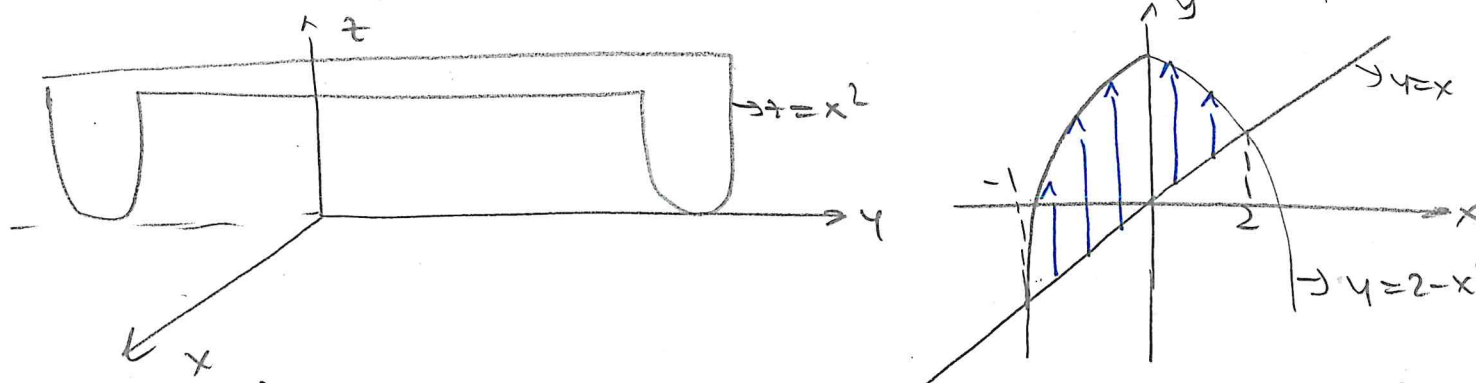
$$\int_0^1 \int_x^{2-x} (x^2 + y^2) dy dx = \int_0^1 \left(x^2 y + \frac{y^3}{3} \right) \Big|_x^{2-x} dx = \int_0^1 \left[x^2(2-x) + \frac{(2-x)^3}{3} - x^3 - \frac{x^3}{3} \right] dx$$

$$= \int_0^1 \left[2x^2 - x^3 + \frac{(2-x)^3}{3} - \frac{4x^3}{3} \right] dx = \frac{2x^3}{3} - \frac{x^4}{4} - \frac{(2-x)^4}{12} \Big|_0^1 = \frac{4}{3}$$

Ex 3 Find the volume of the solid that is bounded above

by the cylinder $z = x^2$ and below by the region enclosed by the parabola $y = 2 - x^2$ and the line $y = x$ in the plane

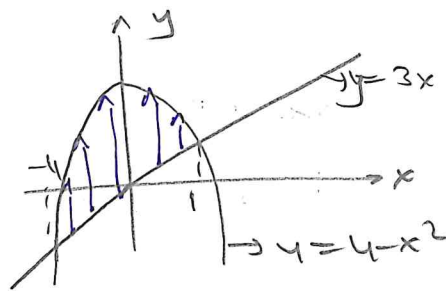
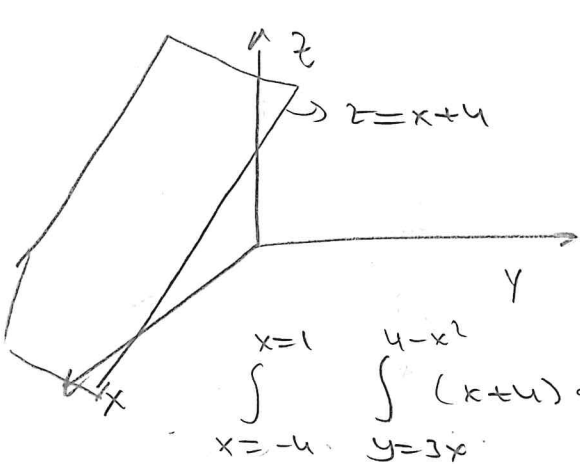
(Another version of question under $z = x^2$ above the region $y = 2 - x^2$ and $y = x$)



$$\int_{-2}^1 \int_x^{2-x^2} x^2 dy dx = \int_{-2}^1 x^2 y \Big|_x^{2-x^2} dx$$

$$= \int_{-2}^1 x^2 (2 - x^2 - x) dx = \frac{2x^3}{3} - \frac{x^5}{5} - \frac{x^4}{4} \Big|_{-2}^1 = \left(\frac{2}{3} - \frac{1}{5} - \frac{1}{4} \right) - \left(-\frac{16}{5} + \frac{32}{5} \right) = \frac{163}{60}$$

Ex 3 Find the volume of the solid whose base is the region in the xy plane that is bounded by the parabola $y=4-x^2$ and the line $y=3x$, while the top of the solid is bounded by the plane $z=x+y$



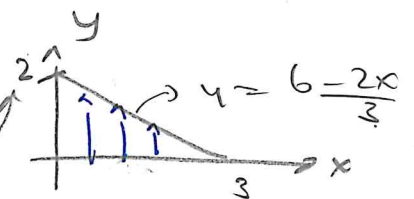
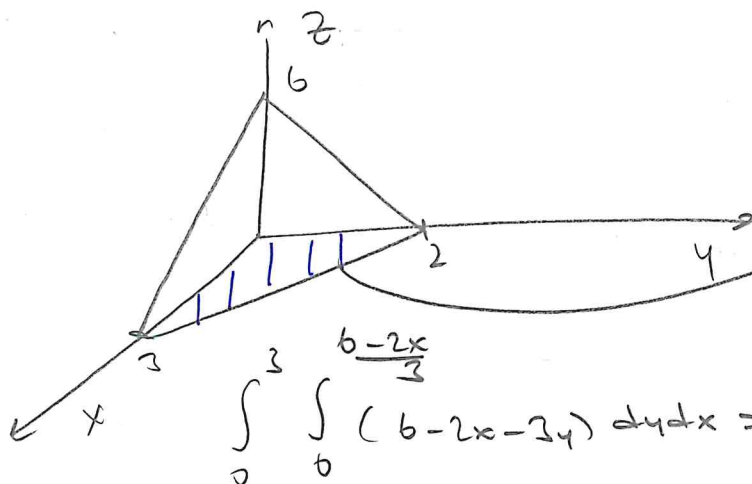
$$\begin{aligned} y &= 3x \wedge y = 4 - x^2 \\ 4 - x^2 &= 3x \\ x^2 + 3x - 4 &= 0 \\ \frac{4}{-1} \\ x &= -4 \quad x = 1 \end{aligned}$$

$$\int_{x=-4}^1 \int_{y=3x}^{4-x^2} (x+y) dy dx = \int_{-4}^1 (x+y)(4-x^2-3x) dx$$

$$= \int_{-4}^1 [x(4-x^2) + 4(4-x^2) - 3x^2 - 12x] dx = \int_{-4}^1 (-x^3 - 7x^2 - 8x + 16) dx$$

$$= -\frac{1}{4}x^4 - \frac{7}{3}x^3 - 4x^2 + 16x \Big|_{-4}^1 = \left(-\frac{1}{4} - \frac{7}{3} + 12\right) - \left(\frac{64}{3} - 64\right) = \frac{625}{12}$$

Ex 4 Find the volume of the region bounded by $z=6-2x-3y$ plane and coordinate axes.

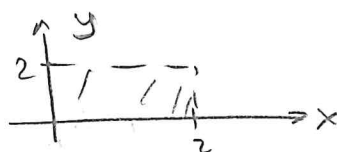
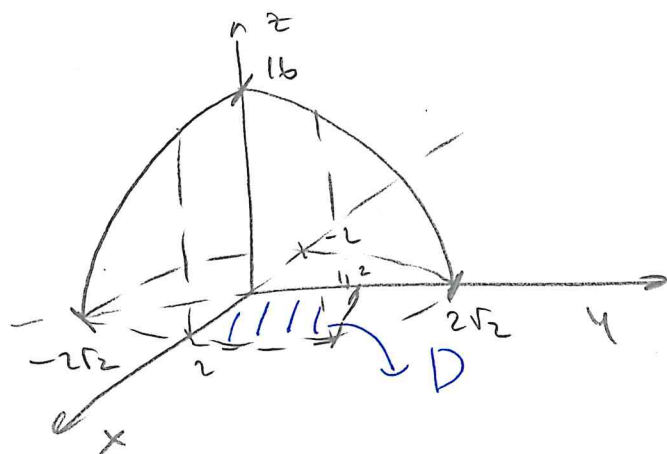


$$\int_0^3 \int_0^{\frac{6-2x}{3}} (6-2x-3y) dy dx = \int_0^3 \left(6y - 2xy - \frac{3y^2}{2}\right) \Big|_0^{\frac{6-2x}{3}} dx$$

$$= \int_0^3 6\left(2 - \frac{2}{3}x\right) - 2x\left(2 - \frac{2}{3}x\right) - 3\left(2 - \frac{2}{3}x\right)^2 dx$$

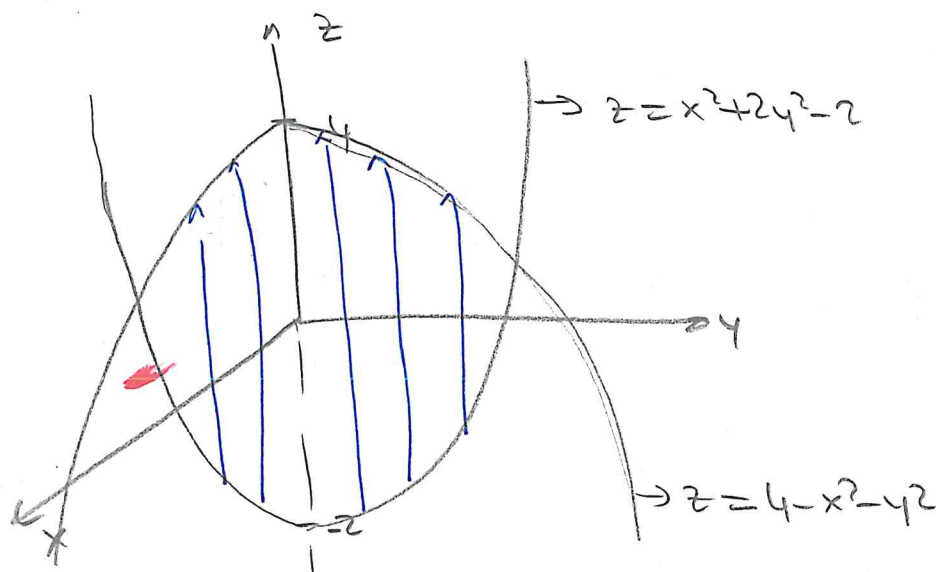
$$= \int_0^3 \left(6 - 4x - \frac{2x^2}{3}\right) dx = \left(6x - 2x^2 - \frac{2}{9}x^3\right) \Big|_0^3 = 6$$

Ex 5 Find the volume of the region bounded by the
 $z = 16 - x^2 - 2y^2$ paraboloid and below by the rectangle
 enclosed by $0 \leq x \leq 2$ $0 \leq y \leq 2$.



$$\begin{aligned}
 & \int_0^2 \int_0^2 (16 - x^2 - 2y^2) dx dy \\
 &= \int_0^2 \left(16x - \frac{x^3}{3} - 2xy^2 \right) \Big|_0^2 dy \\
 &= \int_0^2 \left(32 - \frac{8}{3} - 4y^2 \right) dy \\
 &= \left(32y - \frac{8}{3}y - 4\frac{y^3}{3} \right) \Big|_0^2 \\
 &= 64 - \frac{16}{3} - \frac{32}{3} = 64 - \frac{48}{3} = \frac{48}{3}
 \end{aligned}$$

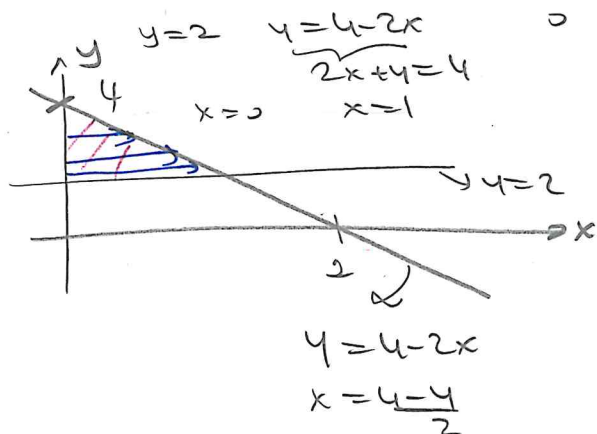
Ex 6 Find volume of the solids inside the paraboloids
 $z = 4 - x^2 - y^2$ and $z = x^2 + 2y^2 - 2$ above the region in the
 xy plane bounded by $x = -1, x = 1, y = -1$ and $y = 1$



$$\begin{aligned}
 & \int_{-1}^1 \int_{-1}^1 \{ (4 - x^2 - y^2) - (x^2 + 2y^2 - 2) \} dy dx = \int_{-1}^1 \int_{-1}^1 (6 - 2x^2 - 3y^2) dy dx \\
 &= \int_{-1}^1 \left(6y - 2x^2y - y^3 \right) \Big|_{-1}^1 dx = \int_{-1}^1 \left[(6 - 2x^2 - 1) - (-6 + 2x^2 + 1) \right] dx \\
 &= \int_{-1}^1 (10 - 4x^2) dx = 10x - \frac{4}{3}x^3 \Big|_{-1}^1 = \frac{52}{3}
 \end{aligned}$$

Ex (Order changing examples) For the following integrals sketch the region of integration, reverse the order of integration and evaluate the integral

① $I = \int_0^1 \int_2^{4-2x} dy dx = ?$



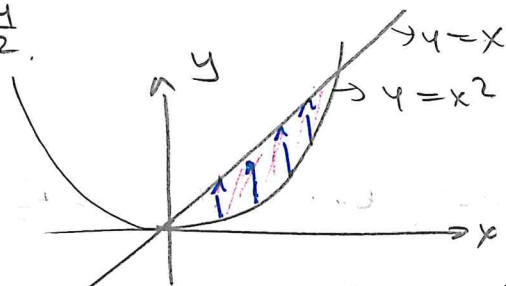
$$\int_0^1 \int_2^{4-2x} dy dx = \int_2^4 \int_0^{(4-y)/2} dx dy$$

$$= \int_2^4 \left(\frac{4-y}{2} \right) dy = 2y - \frac{y^2}{4} \Big|_2^4$$

$$= \left(8 - \frac{16}{4} \right) - \left(4 - \frac{4}{4} \right)$$

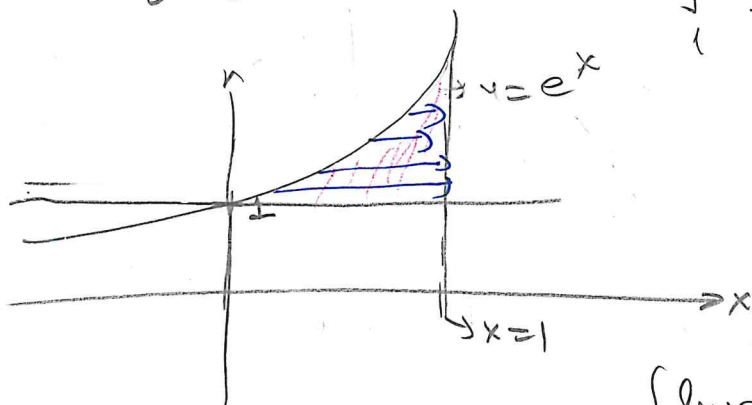
$$= 4 - 3 = 1$$

② $I = \int_0^1 \int_y^{\sqrt{y}} dx dy$



$$I = \int_0^1 \int_{x^2}^x dy dx = \int_0^1 (x - x^2) dx = \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 = \frac{1}{6}$$

③ $I = \int_0^1 \int_1^{e^x} dy dx$



$$\int_0^1 \int_1^{e^x} dy dx = \int_1^e \int_0^{\ln y} dx dy$$

$$= y - (y \ln y - y) \Big|_1^e$$

$$= (2y - y \ln y) \Big|_1^e = (2e - e) - 2$$

$$= e - 2$$

$$\int \ln y dy = \ln y \cdot y - \int \frac{1}{y} \cdot y dy$$

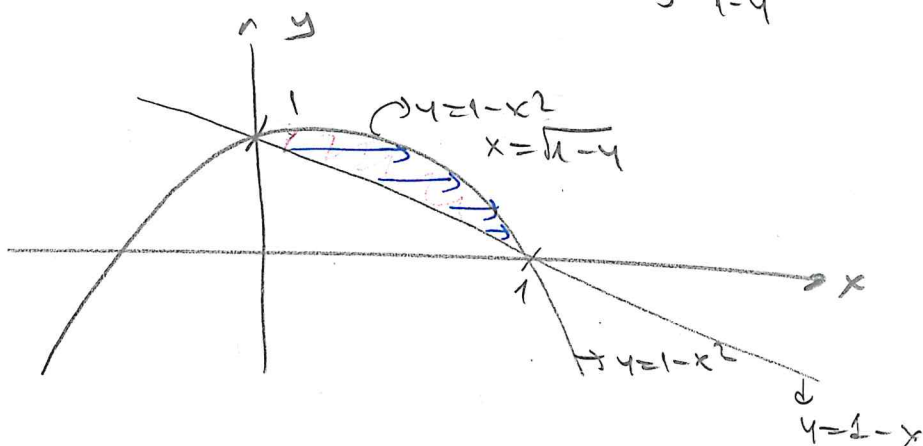
$$\ln y = u \quad \frac{1}{y} dy = du$$

$$dy = du \quad y = v$$

$$\int u \cdot dv = uv - \int v du$$

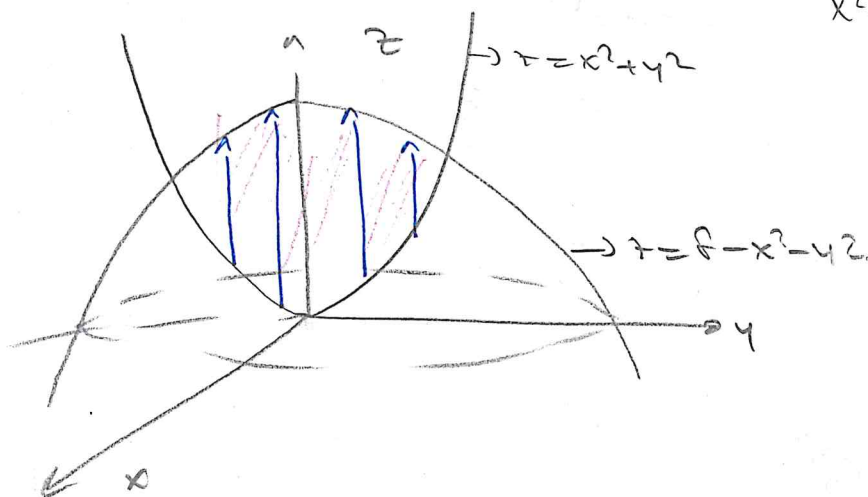
④ $I = \int_0^1 \int_{1-x}^{1-x^2} dy dx = ?$

$I = \int_0^1 \int_{1-y}^{\sqrt{1-y}} dx dy \Rightarrow$ more difficult



$$I = \int_0^1 \int_{1-x}^{1-x^2} dy dx = \int_0^1 ((1-x^2) - (1-x)) dx = \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 = \frac{1}{6}$$

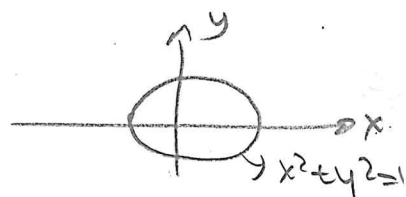
Ex Obtain the volume of the region bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$



$$x^2 + y^2 = z \quad \wedge \quad z = 8 - x^2 - y^2$$

$$2(x^2 + y^2) = 8$$

$$x^2 + y^2 = 4$$



$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (8 - x^2 - y^2) - (x^2 + y^2) dy dx = \int_0^{2\pi} \int_0^2 (8 - 2r^2) r dr d\theta$$

$$= \int_0^{2\pi} \left(4r^2 - \frac{2r^4}{4} \right) \Big|_0^2 d\theta = (16 - 2 \cdot \frac{16}{4}) 2\pi = \boxed{16\pi}$$

DON'T FORGET

CHANGE OF VARIABLES IN DOUBLE INTEGRALS

→ Remember that in Calculus (I) we had a substitution rule that

$$\int_a^b f(g(x))g'(x)dx = \int_c^d f(u)du \quad \begin{array}{l} u=g(x) \Rightarrow x=a \text{ gives } u= \\ x=b \text{ gives } u= \end{array}$$

→ Here we will implement it for double integrals.

→ We say that the transformation is one-to-one from the set S in the uv -plane onto the set D in the xy -plane provided

- (i) every point in S gets mapped to a point in D ,
- (ii) every point in D is the image of a point in S , and
- (iii) different points in S get mapped to different points in D

→ If the transformation is one-to-one, the defining equations can be solved for u and v as functions of x and y and the resulting inverse transformation

$$\begin{array}{l} u = u(x, y) \\ v = v(x, y) \end{array} \quad \text{is one-to-one from } D \text{ onto } S.$$

→ Suppose that the functions $x = x(u, v)$, $y = y(u, v)$ have continuous first partial derivatives and that $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$ at (u, v)

→ Then, the transformation $\begin{array}{l} x = x(u, v) \\ y = y(u, v) \end{array}$ is one-to-one near (u, v)

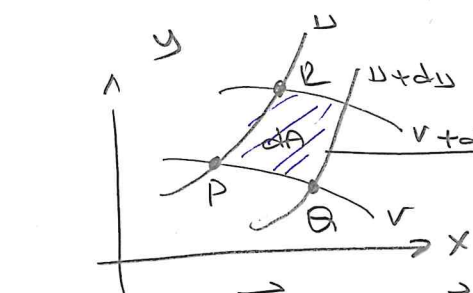
and the inverse transformation has continuous partial derivatives

we also have

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{\frac{\partial(x, y)}{\partial(u, v)}}$$

→ A one-to-one transformation can be used to transform the double integral

$$\iint_D f(x, y) \frac{dA}{dx dy} = \iint_S f(x(u, v), y(u, v)) \frac{dA}{?}$$



Similarly, $\vec{PQ} = \frac{\partial x}{\partial v} dv \vec{i} + \frac{\partial x}{\partial v} dv \vec{j}$

$$dA \approx |\vec{PQ} \times \vec{PR}|$$

$$\vec{PQ} = dx \vec{i} + dy \vec{j}$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$

$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv$$

$\vec{PR} = \frac{\partial x}{\partial u} du \vec{i} + \frac{\partial y}{\partial u} du \vec{j} + \frac{\partial x}{\partial v} dv \vec{i} + \frac{\partial y}{\partial v} dv \vec{j}$
→ res along the curve

Thm Change of variables formula for double integrals

Let $x = x(u, v)$, $y = y(u, v)$ be a one-to-one transformation from a domain S in the uv -plane onto a domain D in the xy -plane. Suppose that the functions x and y , and their first partial derivatives with respect to u and v , are continuous in S . If $f(x, y)$ is integrable on D , and if $g(u, v) = f(x(u, v), y(u, v))$, then g is integrable on S and

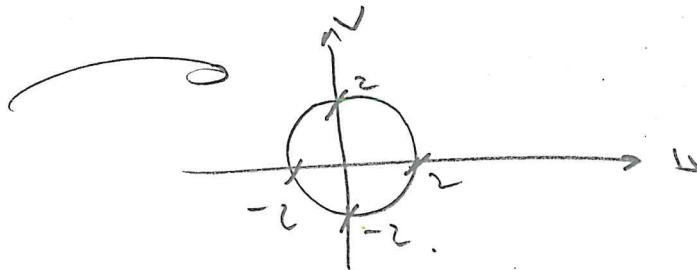
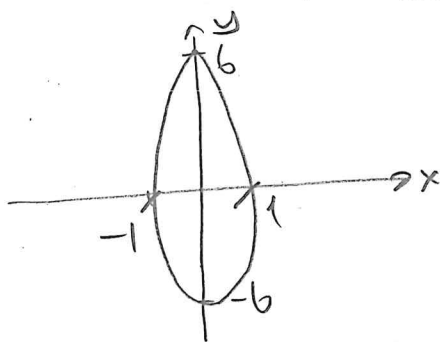
$$\iint_D f(x, y) dx dy = \iint_S g(u, v) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv \quad \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \neq 0$$

$$dA = \begin{vmatrix} \vec{p} & \vec{q} & \vec{k} \\ \frac{\partial x}{\partial u} du & \frac{\partial y}{\partial u} du & 0 \\ \frac{\partial x}{\partial v} dv & \frac{\partial y}{\partial v} dv & 0 \end{vmatrix} = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Ex 1 Determine the new region that we get by applying the given transformation to region R .

① R is the ellipse $x^2 + \frac{y^2}{36} = 1$ and use transformation
 $x = \frac{u}{2}$ $y = 3v$

$$\frac{u^2}{4} + \frac{9v^2}{36} = 1 \Rightarrow u^2 + v^2 = 4 \Rightarrow \text{circle of radius 2}$$



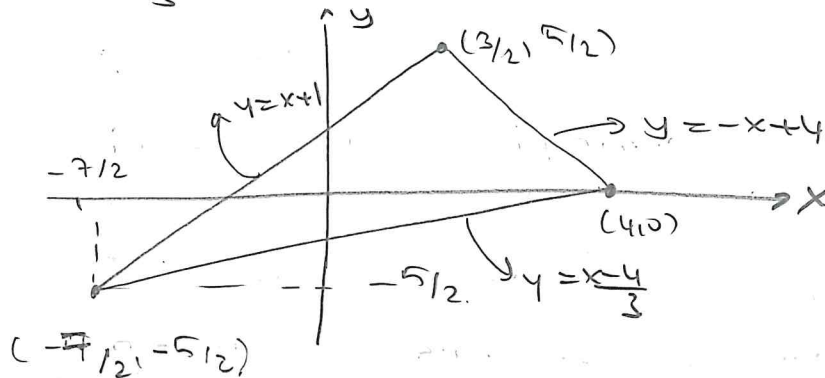
5) R is the bounded by $y = -x + 4$, $y = x + 1$ and $y = \frac{x-4}{3}$ and the transformation is $x = \frac{1}{2}(u+v)$, $y = \frac{1}{2}(u-v)$

Intersection points of lines

$$-x + 4 = x + 1 \Rightarrow x = \frac{3}{2} \left(\frac{3}{2}, \frac{5}{2} \right)$$

$$-x + 4 = \frac{x-4}{3} \Rightarrow -3x + 12 = x - 4 \quad x = 4 \quad (4, 0)$$

$$x + 1 = \frac{x-4}{3} \Rightarrow 3x + 3 = x - 4 \quad x = -\frac{7}{2} \quad \left(-\frac{7}{2}, -\frac{5}{2} \right)$$

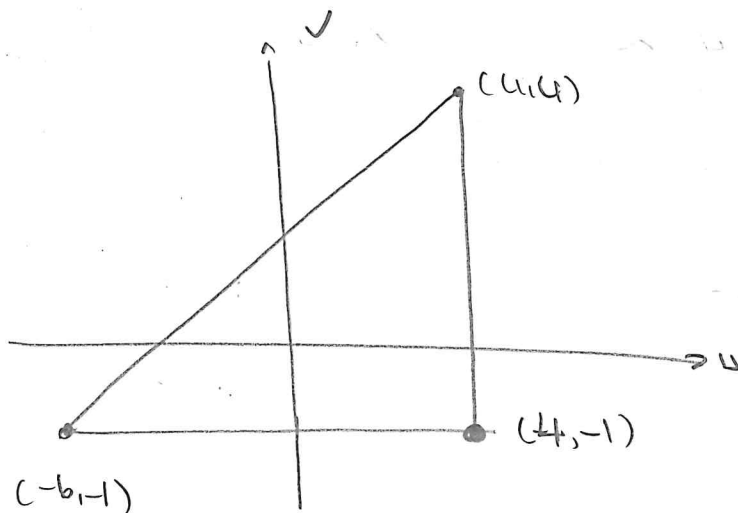


Apply transformation

$$\bullet y = x + 1 \quad \frac{1}{2}(u-v) = \frac{1}{2}(u+v) + 1 \Rightarrow 2v = -2 \quad \boxed{v = -1}$$

$$\bullet y = -x + 4 \quad x = \frac{1}{2}(u+v) \quad y = \frac{1}{2}(u-v) \Rightarrow \frac{1}{2}(u-v) = -\frac{1}{2}(u+v) + 4 \quad \boxed{u = 4}$$

$$\bullet y = \frac{x-4}{3} \quad \frac{1}{2}(u-v) = \frac{1}{6}(u+v) - \frac{4}{3} \Rightarrow 3u - 3v = u + v - 8 \quad \boxed{v = \frac{u}{2} + 2}$$



Intersection Points

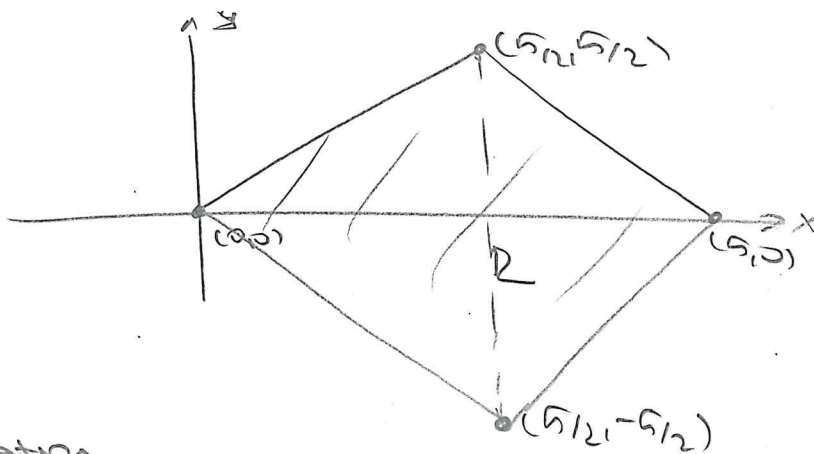
$$v = -1 \quad u = 4 \quad v = \frac{1}{2}u + 2$$

$$-1 = \frac{1}{2}u + 2 \Rightarrow u = -6 \quad (-6, -1)$$

$$v = \frac{1}{2}u + 2 \Rightarrow v = 4 \quad (4, 4)$$

Ex 7 Evaluate $\iint_R (x+y) dA$ where R is the trapezoidal region (shaded yellow) with vertices given by $(0,0)$, $(5,0)$, $(5/2, 5/2)$ and $(5/2, -5/2)$ using the transformation $x=2u+3v$ and $y=2u-3v$

Sketch the region



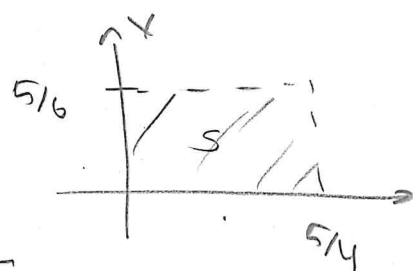
Apply the transformation

• $y=x \Rightarrow 2u-3v = 2u+3v \quad \underline{v=0}$

• $y=-x \Rightarrow 2u-3v = -2u-3v \quad \underline{u=0}$

• $y=-x+5 \Rightarrow 2u-3v = -2u-3v+5 \quad \underline{u=5/4}$

• $y=x-5 \Rightarrow 2u-3v = 2u+3v-5 \quad \underline{v=5/6}$



→ So, the region R becomes a rectangular region S defined by $0 \leq u \leq 5/4$, $0 \leq v \leq 5/6$

$$dA = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv = 12 du dv$$

$$J = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 2 & -3 \end{vmatrix} = -6 - 6 = -12$$

Then, the double integral becomes

$$\iint_R (x+y) dA = \int_0^{5/6} \int_0^{5/4} (2u+3v+2u-3v) 12 du dv$$

$$= \int_0^{5/6} \int_0^{5/4} 4u du dv = \int_0^{5/6} 2u^2 \Big|_0^{5/4} dv$$

$$= 24 \cdot \frac{25}{16} \int_0^{5/6} dv = \frac{125}{4}$$

Ex 7 Evaluate

$$\int_0^4 \int_{x=y/2}^{x=y/2+1} \frac{2x-y}{2} dx dy \quad \text{by applying the}$$

transformation $u = \frac{2x-y}{2}, v = \frac{y}{2}$

So, we have $x = u+v, y = 2v$

Boundaries

$$x = y/2 \Rightarrow u+v = \frac{2v}{2} = v \Rightarrow u = 0$$

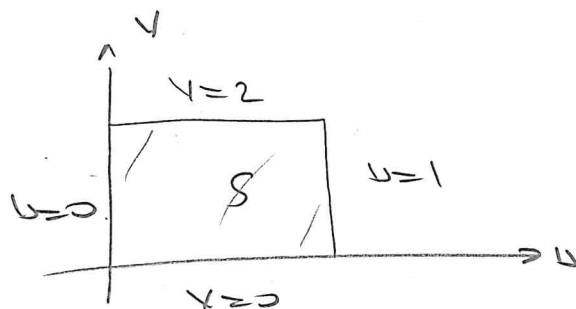
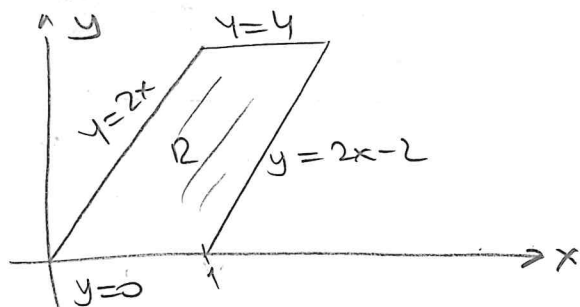
$$x = y/2 + 1 \Rightarrow u+v = \frac{2v}{2} + 1 = v+1 \Rightarrow u = 1$$

$$y = 0 \Rightarrow 2v = 0 \Rightarrow v = 0$$

$$y = 4 \Rightarrow 2v = 4 \Rightarrow v = 2$$

$$|J| = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = 2$$

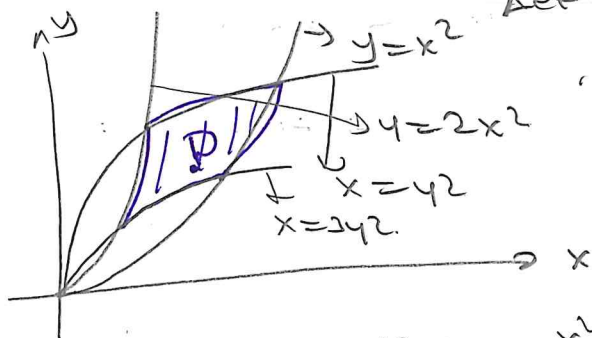
$$\int_0^4 \int_{x=y/2}^{x=y/2+1} \frac{2x-y}{2} dx dy = \int_{v=0}^2 \int_{u=0}^1 u \cdot 2 du dv = \int_{v=0}^2 u^2 \Big|_0^1 dv = 2$$



Ex 7 Find the area of the finite plane region bounded by the

four parabolas $y=x^2, y=2x^2, x=y^2$ and $x=3y^2$

Let choose the change of variable functions $u = \frac{x^2}{y}$ and $v = \frac{y^2}{x}$

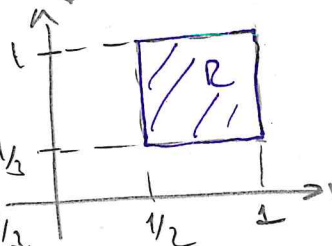


$$y = x^2 \Rightarrow u = 1$$

$$y = 2x^2 \Rightarrow u = 1/2$$

$$x = y^2 \Rightarrow v = 1/3$$

$$x = 3y^2 \Rightarrow v = 1$$



$$\left| \frac{\partial(u,v)}{\partial(x,y)} \right| = \frac{1}{\left| \frac{\partial(x,y)}{\partial(u,v)} \right|} = \frac{1}{3} \quad \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = 1/3$$

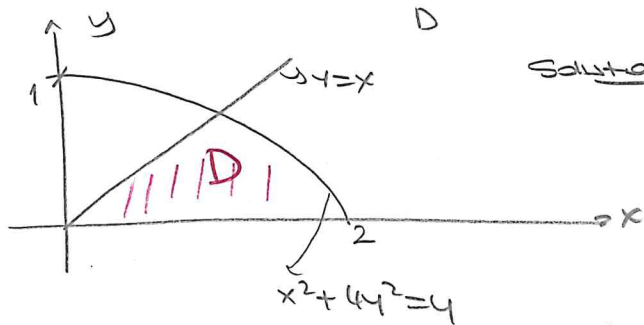
$$\iint_R dx dy = \iint_R \frac{1}{3} du dv = \int_{1/2}^1 \int_{1/3}^1 \frac{1}{3} du dv = \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{9}$$

2

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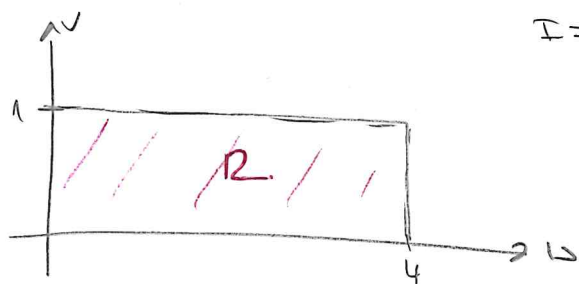
Ex 7 Evaluate $I = \iint_D \frac{y}{x} dx dy$ where D is the following region



Solution Let $u = x^2 + 4y^2$ $v = y/x$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & 8y \\ -y/x^2 & 1/x \end{vmatrix} = 2 + 8 \frac{y^2}{x^2} = 2 + 8v^2$$

$$\text{we need } \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \left| \frac{1}{2 + 8v^2} \right| = \frac{1}{2 + 8v^2}$$



$$y=x \Rightarrow v=1$$

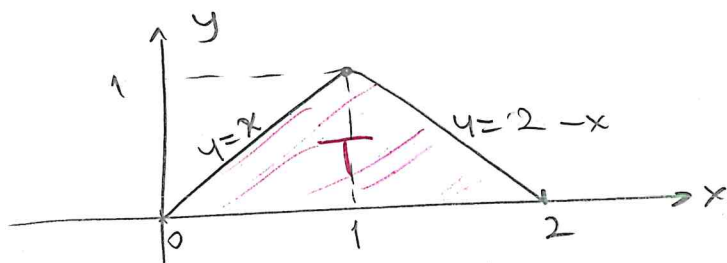
$$x^2 + 4y^2 = 4 \Rightarrow u=4$$

$$x\text{-axis} \Rightarrow v=0$$

$$(0,0) \Rightarrow u=0$$

$$\begin{aligned} I &= \iint_R \frac{v}{2 + 8v^2} du dv = \int_0^1 \int_2^{10} \frac{v}{2 + 8v^2} du dv \\ &= \int_0^1 \left[\frac{du}{16} \right]_2^{10} dv = \int_0^1 \frac{1}{16} \ln \frac{u}{2} dv \\ &= \frac{1}{16} (\ln 10 - \ln 2) \cdot 1 = \frac{1}{16} \ln 5 \end{aligned}$$

Ex Evaluate $I = \iint_T (x+y)^3 dx dy$ over the triangle T with vertices $(0,0)$, $(1,1)$ and $(2,0)$



$$u = y-x, \quad v = y+x$$

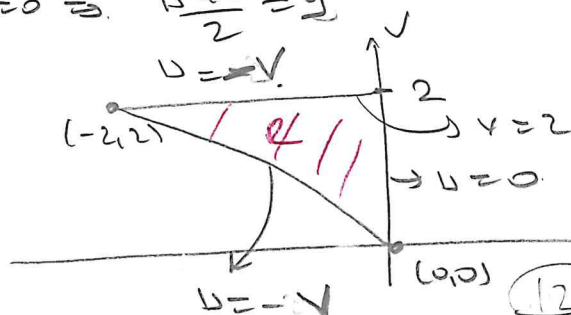
$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2$$

$$dx dy = \frac{1}{\left| \frac{\partial(u,v)}{\partial(x,y)} \right|} du dv = \frac{1}{2} du dv$$

$$y=x \Rightarrow x-y=0 \Rightarrow u=0$$

$$y=2-x \Rightarrow y+x=2 \Rightarrow v=2$$

$$x\text{-axis } y=0 \Rightarrow \frac{u+v}{2} = 0 \Rightarrow u = -v$$



$$I = \int_{-2}^2 \int_0^2 \frac{v^3}{2} du dv$$

$$= \int_{-2}^2 v \frac{v^3}{2} dv = \int_{-2}^2 \frac{v^4}{2} dv$$

$$= \frac{v^5}{10} \Big|_{-2}^2 = \frac{16}{5}$$

